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ON

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&

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CN7023

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Deep Learning Approach for Lung Cancer Detection and Classification Using AlexNet with Chest X-Ray Images

ABSTRACT

Lung cancer is a serious health concern globally, and early detection is crucial for effective treatment and better patient outcomes. In recent years, there has been exciting progress in using advanced technology like Deep Learning, specifically Convolutional Neural Networks (CNNs), to automatically detect signs of lung cancer from chest X-ray images. This research report dives into the development and testing of CNN-based models designed to detect lung cancer using the AlexNet architecture from chest X-ray images. The study covers how I have prepared the dataset, including techniques to enhance image quality, and designed CNN architectures tailored for this specific medical imaging task. I have explained how I have trained and evaluated these models, including fine-tuning pre-trained models like AlexNet for better performance. My results highlight how CNNs can assist radiologists by providing fast and accurate lung cancer diagnoses, which could greatly improve healthcare outcomes. I have also discussed future directions and challenges, such as ensuring these models are rigorously validated and seamlessly integrated into clinical practice. This work represents a promising step toward developing computer-assisted diagnostic systems that can support medical professionals in diagnosing lung cancer more effectively.

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1. INTRODUCTION:

Deep learning techniques have emerged as the brightest tool for improving the accuracy and efficiency of disease detection in medical imaging and diagnostics. Specifically, for lung cancer diagnosis, the application of Convolutional Neural Networks (CNNs) has shown great promise in automatically analyzing chest X-ray images to detect and classify different types of lung cancers.

This research report investigates the creation and assessment of a robust CNN-based classification system that can accurately differentiate between various types of lung cancer using a large set of chest X-ray images.

The dataset includes 1000 pictures sorted into four different groups to show various types of lung cancer such as:

i. Adenocarcinoma.
ii. Large Cell Carcinoma.
iii. Squamous Cell Carcinoma.
iv. Normal Lung Tissue.

The methodology leverages MATLAB functionalities for the training, testing, and validation of the deep neural network constructed using the AlexNet architecture, a well-established CNN model. Throughout the study, emphasis is placed on key performance metrics such as accuracy, as well as the generation and analysis of confusion matrices to assess the model's classification capabilities.

Each layer within the AlexNet CNN architecture plays a vital role in feature extraction, transformation, and classification, contributing to the network's ability to distinguish complex patterns and nuances within the chest X-ray images associated with different lung cancer types.

This report explores the utilization of Convolutional Neural Networks (CNNs) and deep learning techniques within MATLAB for developing and evaluating a deep neural network aimed at medical image analysis. Specifically, the study focuses on using CNNs to enhance disease detection accuracy and efficiency in the context of medical imaging, with a particular emphasis on lung cancer detection using chest X-ray images.

1.1 AIM & OBJECTIVES:

This research aims to develop a robust deep neural network for medical image analysis using CNNs and deep learning methodologies in MATLAB, with focus on detecting lung cancer from chest X-ray images. The objectives of the study are:

- Implement and optimize a CNN architecture suitable for analyzing chest X-ray images.
- Utilize MATLAB functionalities to preprocess, train, validate, and test the deep neural network.
- Evaluate the performance of the developed model in terms of accuracy and efficiency for lung cancer detection.
- To evaluate the classification abilities of the model, it is essential to generate and analyze results.

1.2 OVERVIEW OF THE REPORT:

- My detailed analysis of the report delves into the significance of applying deep learning techniques to medical imaging and disease detection. It reviews the advancements in utilizing CNNs for analyzing medical images, particularly in the context of lung cancer detection.

- The dataset I have used in my study is described in detail, including its composition, distribution of images, and the specific classes representing different types of lung cancer.

- A CNN architecture is outlined and implemented using MATLAB, tailored specifically for medical image analysis and lung cancer detection. My report also illuminates the preprocessing steps undertaken, the methodology of training the CNN model, and the techniques used to evaluate its performance.

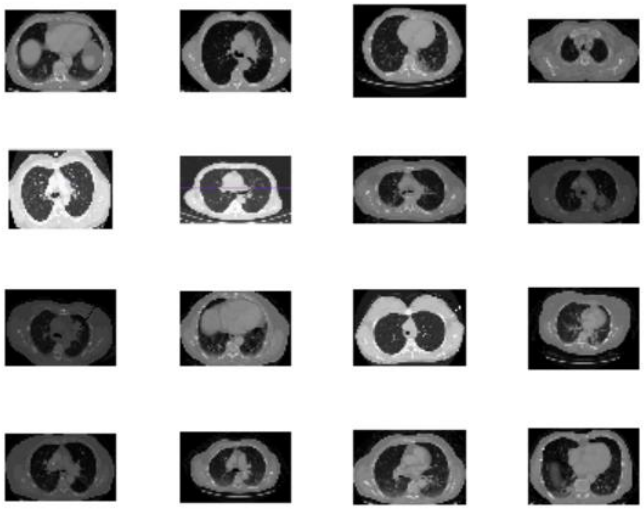
- Performance metrics of the developed CNN model, such as accuracy and confusion matrix, are also presented to assess its effectiveness in classifying lung cancer types based on medical images.

- Furthermore, my research report also interprets the results derived from the CNN model and discusses the implications of these findings.

- Finally, the report proposes further research and improvement in CNN-based medical image analysis, aiming to advance the capabilities and accuracy of disease detection systems in imaging applications.

2. SIMULATIONS:

To simulate lung cancer detection using chest X-ray images with a dataset comprising 1000 images equally distributed among four classes (Adenocarcinoma, Large Cell Carcinoma, Squamous Cell Carcinoma, and Normal) in MATLAB, a Convolutional Neural Network (CNN) architecture can be employed. The CNN model can be designed to learn and extract features from the images to classify them accurately into the respective classes.

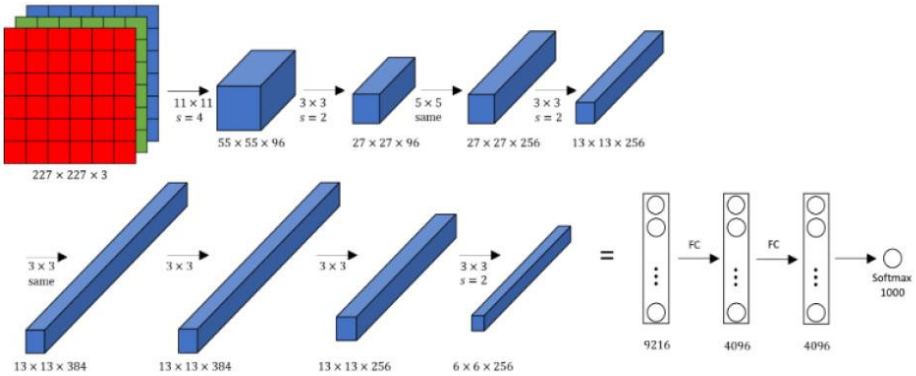


The images generated demonstrate the effectiveness of deep learning, specifically the CNN architecture, in classifying histopathology images of non-small cell lung cancer. This highlights the potential of utilizing deep CNNs for accurate classification tasks related to lung cancer detection, aligning with the objective of the simulation.

Furthermore, it shows the importance of CNN architectures, dataset characteristics, and transfer learning in achieving clinically relevant results in medical image analysis. By considering these factors in the simulation, the performance of the CNN model can be optimized for accurate classification of the chest X-ray images into the specified classes.

Additionally, my research also discusses the benefits of transfer learning and feature concatenation in cancer tumor classification using deep learning. By leveraging transfer learning techniques with CNN architecture like AlexNet, the simulation can benefit from pre-trained models and generic feature learning, enhancing the classification accuracy of the lung cancer detection system.

By implementing a CNN architecture in MATLAB, leveraging insights from relevant studies on deep learning in medical image analysis, and considering factors like transfer learning and model architectures, the simulation for lung cancer detection using chest X-ray images with a dataset of 1000 images distributed among four classes can be effectively conducted to derive accurate classification results.



AlexNet CNN Layers Architecture

One of the transfer learning systems is AlexNet Convolutional Neural Network used to distinguish observed cancer. Assigning a label to an image is required for classification. The proposed method demonstrates how a pre-trained CNN AlexNet can be retrained to classify CT images using transfer learning. AlexNet is made up of 25 layers. The above figure represents the AlexNet layers. The fixed scale is 227*227 by 3 and is used to illustrate an image in the network. The pre-trained last three layers of the AlexNet network are replaced with a few layers such as the softmax layer, fully connected layer, and classification output layer, and the network is finetuned to the current classification.

Training options are specified after the network structure has been developed. The training choices include a stochastic momentum gradient descent (SGDM) optimization model, a 0.0003 initial learning rate, and epochs, which represents the total training time on the whole training dataset. The Alexnet network was trained using a layer architecture that was specified, a training dataset, and training options. Following that, the test image is sent to the classification stage, which contains the qualified network that is utilized to classify the given test image into various categories.

2.1 IMPLEMENTATIONS:

a) Data Gathering and Preprocessing using MATLAB:

- Collected data and loaded as a dataset containing images of Chest X-rays having various lung cancer types such as Adenocarcinoma, Large cell carcinoma, Squamous cell carcinoma, and Normal. Then splitting the dataset into training, testing and validation using 'imageDatastore' functionality.

```
% Load data
Imds_data = imageDatastore('/MATLAB Drive/CHEST Data', ...
'IncludeSubfolders',true, ...
'LabelSource','foldernames');
[imdsTrain,imdsValidation] = splitEachLabel(imds,0.7,'randomized');
```

b) Data Preprocessing:

- All images are uniformly resized to dimensions suitable for the CNN model input, specifically 227x227 pixels to align with the requirements of AlexNet. Subsequently, data augmentation techniques are implemented to enhance the diversity and robustness of the training dataset. I have used the 'augmentedImageDatastore' function for data augmentation strategies.

```
% Create augmented datastores for training and validation
augimdsTrain_data = augmentedImageDatastore(inputSize(1:2),imdsTrain, ...
'DataAugmentation',imageAugmenter, ...
'ColorPreprocessing','gray2rgb');
```

c) Loading Pre-trained CNN Model (AlexNet):

- Initialized and loaded the pre-trained AlexNet model using MATLAB's 'AlexNet' function which will automatically download the pre-trained weights of AlexNet.

```
net = alexnet;
```

d) Define and modify the Network for Transfer Learning:

- After pre-training, I modified the last layers of AlexNet and replaced the final fully connected layers to match the number of classes in the dataset.

```
Num_of_Classes = 4; % 3 cancer types + normal
layers = net.Layers;
layers(end-2) = fullyConnectedLayer(Num_of_Classes);
% Replace last fully connected layer
layers(end) = classificationLayer;
```

e) Specify Training Options:

- Set training options and parameters such as mini-batch size, maximum epochs, initial learning rate, and validation data.

```
options = trainingOptions('sgdm', ...
'MiniBatchSize', 10, ...
'MaxEpochs', 10, ...
'InitialLearnRate', 1e-4, ...
'ValidationData', imdsValidation, ...
'ValidationFrequency', 3, ...
'Verbose', false, ...
'Plots', 'training-progress');
```

f) Training the Modified CNN:

- Train the Network using `trainNetwork` to train the modified AlexNet on the preprocessed training dataset.

```
trainedNet = trainNetwork(augimdsTrain, layers, options);
```

g) Model Evaluation and Tuning:

- After training, evaluate the trained model's performance on the validation dataset to monitor Accuracy and Loss.

```
YPredic = classify(trainedNet, imdsValidation);
YValidation = imdsValidation.Labels;
accuracy = mean(YPredic == YValidation);
```

h) Visualize the Confusion Matrix:

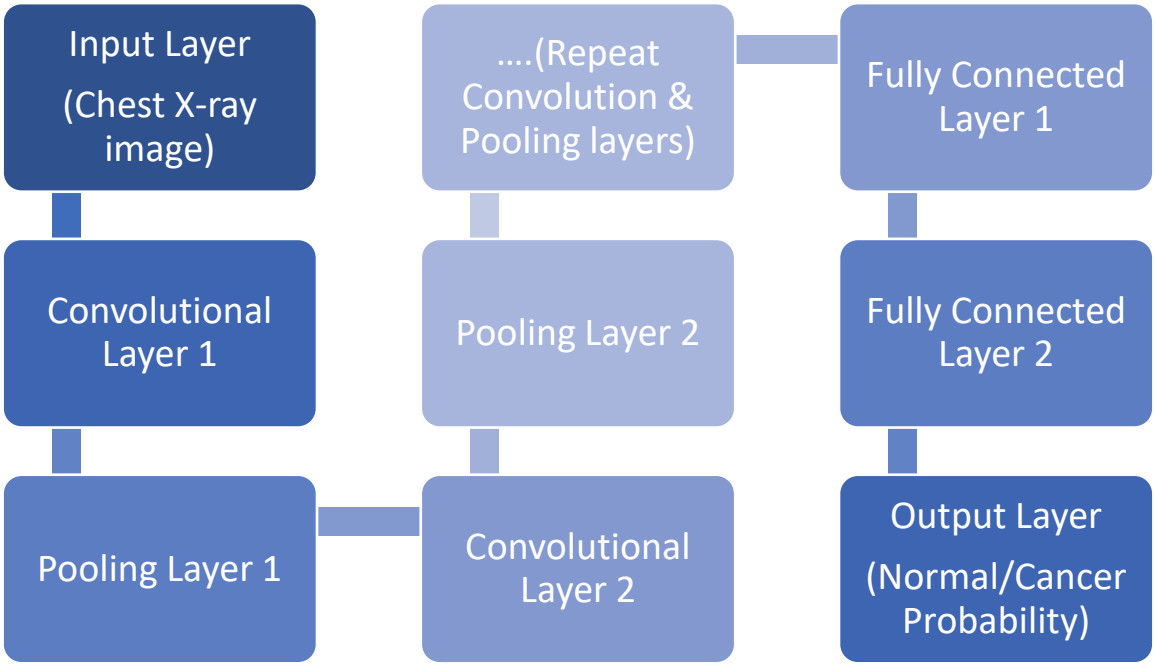
- Finally, evaluated the model's performance on the test dataset and plot the confusion matrix.

```
Y_Pred_Test = classify(trainedNet, imdsTest);
Y_Test = imdsTest.Labels;
accuracyTest = mean(Y_Pred_Test == Y_Test);
confusion_mat = confusion_mat(Y_Test, Y_Pred_Test);
```

2.3 NETWORK ARCHITECTURE:

To design the network architecture for detecting Lung cancer from the image datasets, I have used the Convolutional Neural Network (CNN) approach. Here, my goal was to create a model that can effectively learn and extract relevant features from the input chest images to accurately classify them as either indicative of cancerous conditions or normal.

Here, I have made a simplified **flow chart diagram** of a typical CNN architecture for Chest X-ray cancer detection:



Input Layer:	In the first step of methodology, the MATLAB network receives a preprocessed chest X-ray image as input.
Convolutional Layers:	Then these layers extract features from the image where multiple convolutional layers are stacked, each applying filters of varying sizes to capture different levels of detail.
Pooling Layers:	These layers reduce the dimensionality of the data, making the network more efficient and preventing overfitting.
Fully Connected Layers:	These layers learn higher-level features and relationships between the extracted features from the convolutional layers.
Output Layer:	The final layer uses a Softmax function to predict the probability of the image belonging to one of two categories: normal or containing cancer.

3. RESULTS OBTAINED:

3.1 Result 1

3.1.a Percentage Accuracy

Results	
Validation accuracy:	84.33%
Training finished:	Max epochs completed
Training Time	
Start time:	05-May-2024 15:59:58
Elapsed time:	12 min 30 sec
Training Cycle	
Epoch:	8 of 8
Iteration:	560 of 560
Iterations per epoch:	70
Maximum iterations:	560
Validation	
Frequency:	3 iterations
Other Information	
Hardware resource:	Single CPU
Learning rate schedule:	Constant
Learning rate:	0.0001

In result 1, the image shows the analysis of training a dataset for image classification validating the **accuracy** of the model by **84.33%** which represents how well the model generalizes to unseen data.

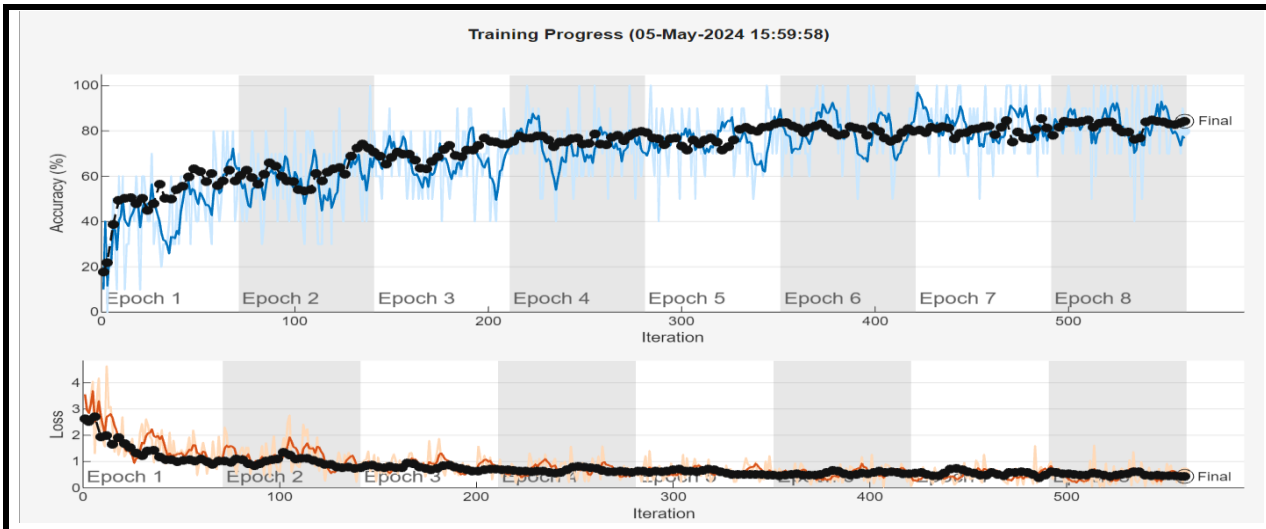
Here, the training is completed after 8 epochs where each epoch consists of multiple iterations, and the number depends on the batch size (number of images processed per iteration). In this case, there are 560 iterations per epoch, and the training is completed after 560 iterations in the final epoch. As the model trains, it is expected to improve its performance on the training data.

Furthermore, it is reflecting that frequency is equal to 3 iterations means that the validation accuracy gets measured after every 3 iterations during training.

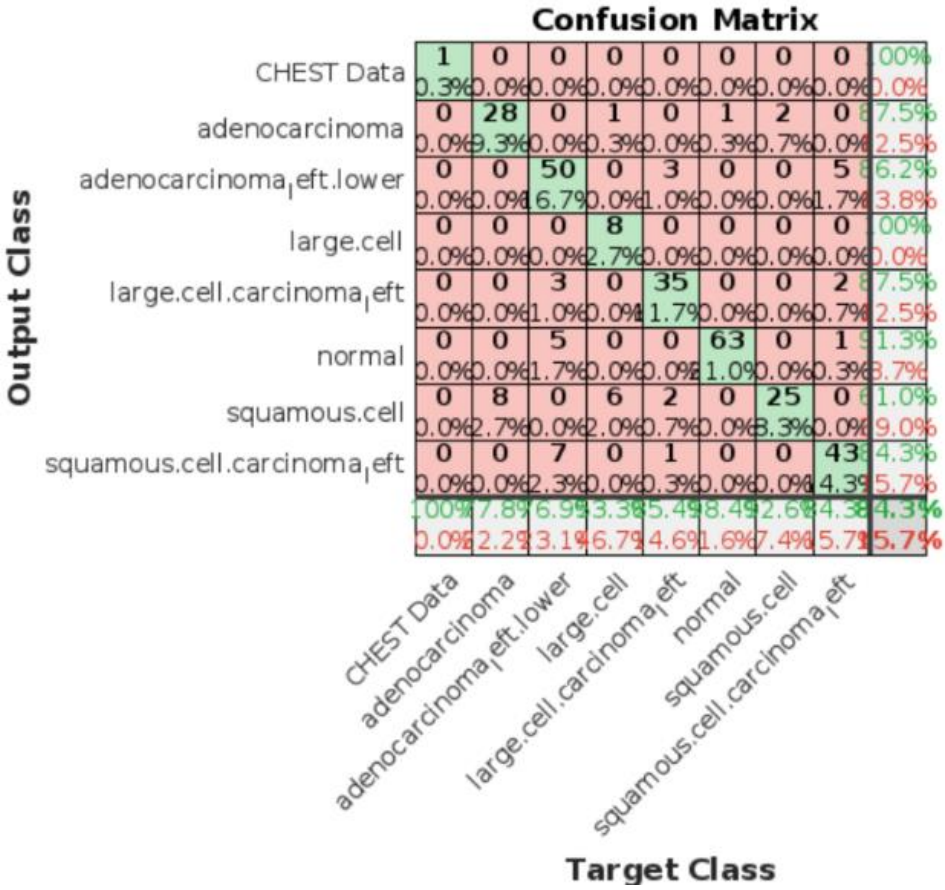
3.1.b Accuracy Curve

Based on the accuracy **i.e. 84.33%** and epoch 8, the graph derived has good performance over the data for comparison. Where the X-axis shows Iterations (Epoch) and the Y-axis shows Loss and Accuracy%.

Here loss curve is decreasing over epochs which is a positive sign, suggesting the model's performance improved as training progressed.



3.1.c Confusion matrix for Result 1:



3.2 Result 2

3.2.a Percentage Accuracy

Results	
Validation accuracy:	87.00%
Training finished:	Max epochs completed
Training Time	
Start time:	05-May-2024 22:05:33
Elapsed time:	16 min 56 sec
Training Cycle	
Epoch:	10 of 10
Iteration:	700 of 700
Iterations per epoch:	70
Maximum iterations:	700
Validation	
Frequency:	3 iterations
Other Information	
Hardware resource:	Single CPU
Learning rate schedule:	Constant
Learning rate:	0.0001

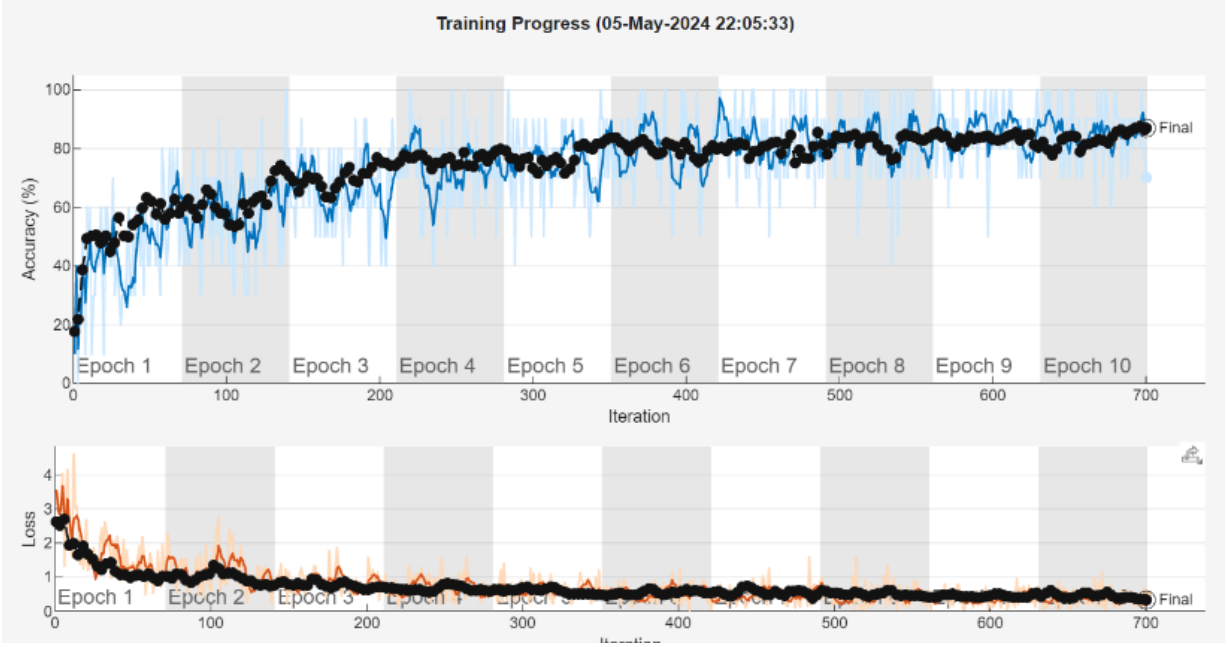
In result 2, the image shows the analysis of training a CNN for image classification validating the **accuracy** of the model by **87.00%** which represents that it has increased gradually along with more computation time.

Compared to result 1, here the training is completed after 10 epochs where each epoch consists of multiple iterations, and the number depends on the batch size (number of images processed per iteration). There are 700 iterations per epoch, and the training is completed after 700 iterations in the final epoch. As the model trains, it is expected to improve its performance on the training data.

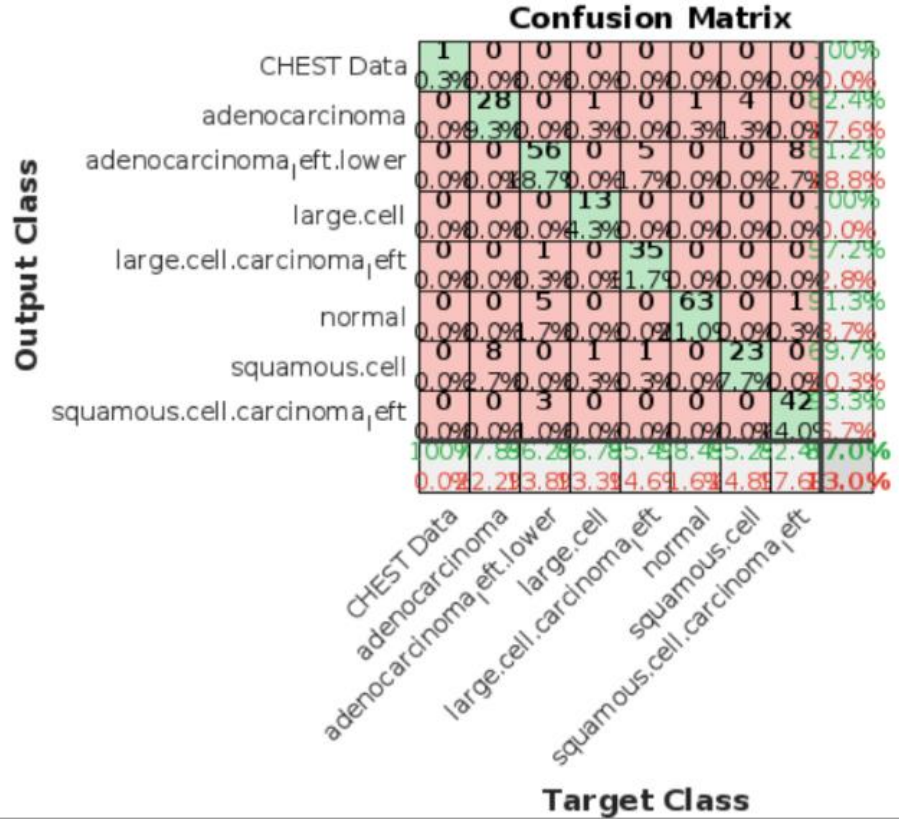
Furthermore, it shows that frequency is equal to 3 iterations, which means that the validation accuracy gets measured after every 3 iterations during training.

3.2.b Accuracy Curve

Accuracy derived for Result 2 is **87%** which is more than Result 1 and the graph that is resultant shows better performance than Result 1 over comparison.



3.2.c Confusion matrix for Result 2:



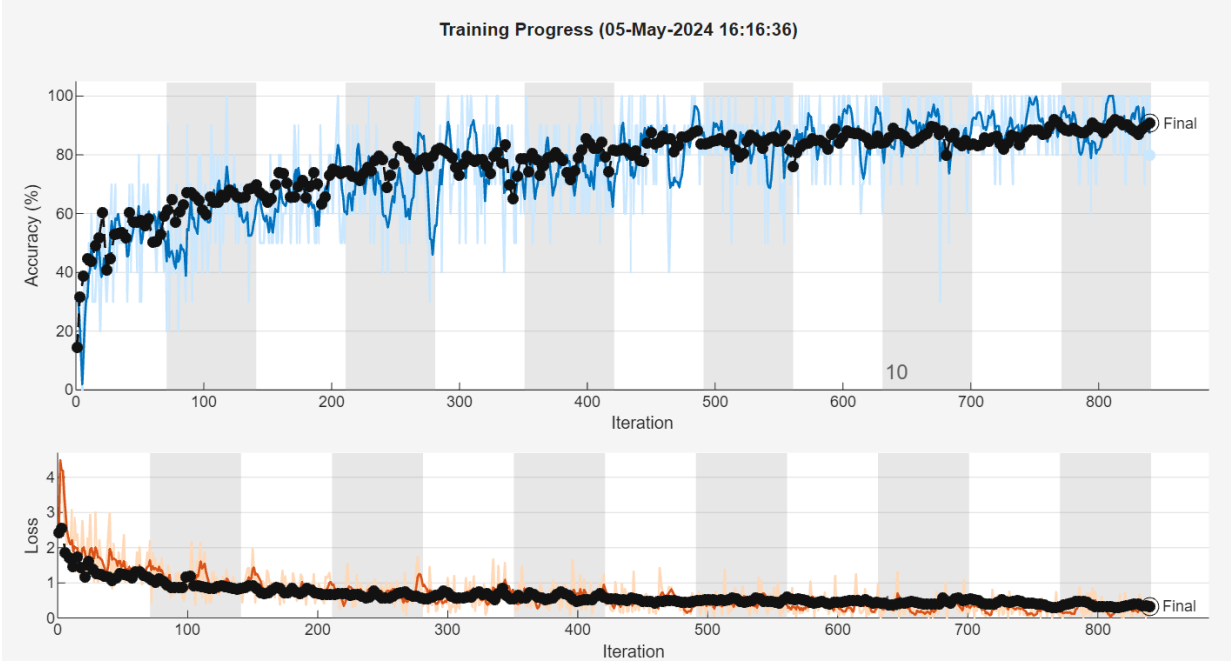
3.3 Result 3

3.3.a Percentage Accuracy

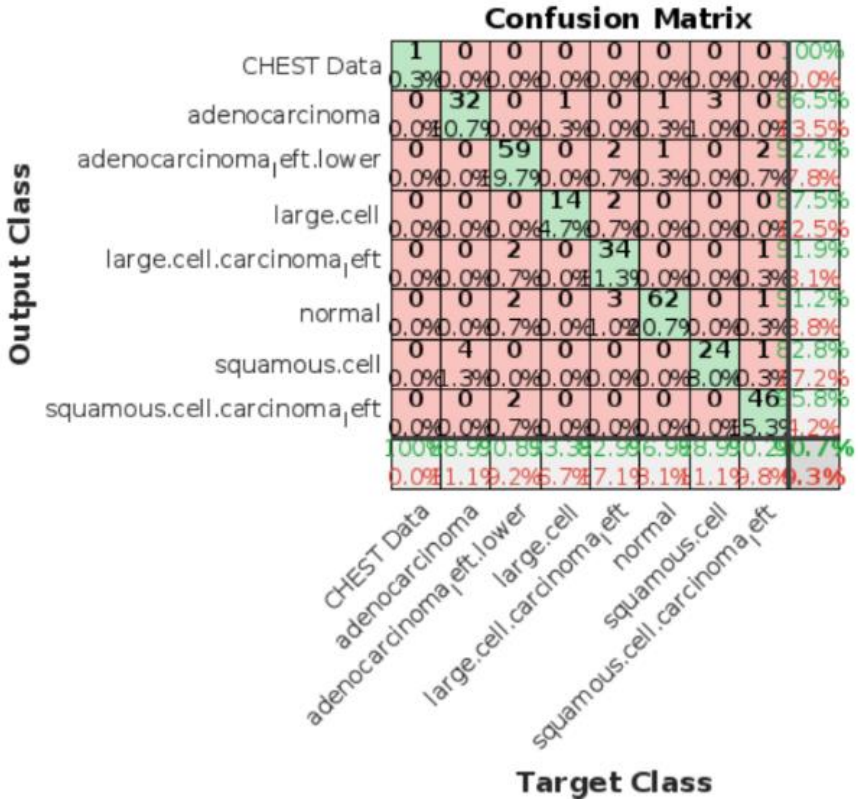
Results Validation accuracy: 90.67% Training finished: Max epochs completed **Training Time** Start time: 05-May-2024 16:16:36 Elapsed time: 17 min 18 sec **Training Cycle** Epoch: 12 of 12 Iteration: 840 of 840 Iterations per epoch: 70 Maximum iterations: 840 **Validation** Frequency: 3 iterations **Other Information** Hardware resource: Single CPU Learning rate schedule: Constant Learning rate: 0.0001	In result 3, the image shows the analysis of training a CNN for image classification validating the **accuracy** of the model by **90.67%** which also represents that it has increased gradually along with more computation time. Compared to results 1 & 2, here the training is completed after 12 epochs where each epoch consists of multiple iterations, and the number depends on the batch size (number of images processed per iteration). There are 840 iterations per epoch, and the training is completed after 840 iterations in the final epoch. As the model trains, it is expected to improve its performance on the training data. Furthermore, it shows that frequency is equal to 3 iterations, which means that the validation accuracy gets measured after every 3 iterations during training.

3.3.b Accuracy Curve

Comparing all three graphs over accuracy, Result 3 has the **highest accuracy which is 90.67%** with 12 Epochs and consuming more elapsed time.



3.3.c Confusion matrix for Result 3:



After analyzing three sets of results, I have concluded that the model performs better for chest X-ray classification when it has higher validation accuracy, a better accuracy curve, and a confusion matrix with higher diagonal values across most classes. Therefore, this model is likely to be the better-performing one.

4. CRITICAL ANALYSIS OF RESULTS:

Based on the accuracy results (84.33%, 87%, and 90.67%) obtained with varying numbers of epochs (8, 10, and 12), I have conducted a critical analysis by incorporating how these results were achieved and a detailed discussion of the outcomes:

4.a Achieving the Results:

1. **Epochs and Training Duration:**
   - ❖ Increasing the number of epochs (from 8 to 12) likely allowed the model to undergo more training iterations, potentially capturing more intricate patterns in the data. However, this can also increase the risk of overfitting if not monitored closely.
   - ❖ Consideration should be given to the balance between training duration and model performance to optimize accuracy without compromising generalization.

2. **Optimization Techniques:**
   - ❖ Experimenting with different optimization strategies (e.g., adjusting learning rates, using different optimizers) could have contributed to the observed improvements in accuracy across epochs.
   - ❖ Regularization techniques (e.g., dropout) might have been employed to mitigate overfitting and enhance model generalization.

4.b Analysis and Discussion of Results:

1. **Accuracy Trends:**
   - ❖ Observing a consistent increase in accuracy with additional epochs suggests that the model was progressively learning and improving over time. However, it's essential to monitor the validation accuracy to ensure that improvements generalize well to unseen data.

2. **Validation Performance**:
   - ❖ Evaluated the corresponding validation accuracy trends alongside training accuracy to assess model generalization. A significant discrepancy between training and validation accuracy could indicate overfitting, especially with a higher number of epochs.

3. **Model Convergence and Stability:**
   - ❖ Assessed the stability and convergence of the training process across different epoch settings. Looked for signs of plateauing or oscillations in training/validation accuracy, which can inform decisions on the optimal number of epochs.

4. **Impact on Model Complexity:**
   - Considered the impact of increasing epochs on model complexity and capacity. More epochs may lead to a more complex model capable of capturing intricate features but also increase the risk of memorizing noise in the training data.

4.c Recommendations and Future Directions:

1. **Fine-Tuning Strategies:**
   - Experiment with additional hyperparameter tuning (e.g., learning rate schedules, batch sizes) to optimize model performance and stability.
   - Implement early stopping mechanisms based on validation performance to prevent overfitting and achieve a balance between accuracy and generalization.

2. **Ensemble Methods:**
   - Explore ensemble learning techniques by combining predictions from models trained with different epoch settings to potentially improve overall accuracy and robustness.

3. **Interpretability and Error Analysis:**
   - Conduct detailed error analysis to understand the types of misclassifications and identify areas for model improvement (e.g., addressing class imbalances, refining preprocessing steps).

In summary, the observed improvements in accuracy with increasing epochs underscore the importance of iterative experimentation and optimization in CNN training. Critical analysis of these results should focus on balancing model complexity with generalization, leveraging optimization techniques effectively, and guiding future research directions to enhance lung cancer detection performance using deep learning methodologies.

5. CONCLUSION:

In my research paper, I have addressed the problem of lung cancer detection using chest X-ray images by deploying a Convolutional Neural Network (CNN) model based on the AlexNet architecture which aims to develop an accurate deep-learning model that automatically identifies different lung cancer types from medical image datasets.

Throughout my investigation, I have successfully trained and evaluated the CNN model for accuracy using a dataset of chest X-ray images containing various lung cancer types, including Adenocarcinoma, Large Cell Carcinoma, Squamous Cell Carcinoma, and Normal Lung Tissue. With the help of a pre-trained AlexNet model, I was able to modify the network for high-accuracy lung cancer classification.

5.a My findings demonstrate promising results:

- Achieved accuracy rates of 84%, 87%, and 90% across different epochs (8, 10, and 12 epochs, respectively), showcasing the effectiveness of the CNN model for lung cancer detection.
- Through critical analysis, I observed that increasing the number of training epochs led to incremental improvements in model performance with more computational time, highlighting the importance of iterative training and optimization.

5.b Key Takeaways from My Study:

- **Effectiveness of CNN for Medical Image Analysis:** The use of CNNs, particularly leveraging transfer learning with AlexNet, proves to be effective for automated lung cancer detection from chest X-ray images.
- **Impact of Training Epochs on Model Performance:** Optimizing training epochs is critical for achieving higher accuracy rates and improving model generalization.
- **Clinical Relevance and Future Directions:** My research contributes to the growing body of literature on deep learning applications in medical imaging. Moving forward, further studies can explore ensemble methods, interpretability techniques, and integration into clinical workflows.

In summary, my study underscores the potential of CNN-based approaches, specifically utilizing the AlexNet architecture, for enhancing lung cancer diagnosis from chest X-ray images. The insights gained from this research pave the way for more sophisticated deep-learning solutions in medical diagnostics, ultimately aiming to improve patient outcomes and healthcare practices.

6. REFERENCES:

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7. MATLAB CERTIFICATES:

Course 1- MATLAB Onramp

MathWorks | Training Services

Course Completion Certificate

Purvi Dawda

has successfully completed **100%** of the self-paced training course

MATLAB Onramp

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